Journal of King Saud University – Computer and Information Sciences 34 (2022) 6365–6376

Contents lists available at ScienceDirect

Journal of King Saud University – Computer and Information Sciences

journal homepage: www.sciencedirect.com

An adaptive outlier removal aided k-means clustering algorithm

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ARTICLE INFO

Article history:
Received 1 April 2021
Revised 9 June 2021
Accepted 2 July 2021
Available online 13 July 2021

Keywords:
Data clustering
Modified k-means
Outliers removal
Adapted Tukey's rule
New distance metric

ABSTRACT

K-means is one of ten popular clustering algorithms. However, k-means performs poorly due to the presence of outliers in real datasets. Besides, a different distance metric makes a variation in data clustering accuracy. Improve the clustering accuracy of k-means is still an active topic among researchers of the data clustering community from outliers removal and distance metrics perspectives. Herein, a novel modification of the k-means algorithm is proposed based on Tukey's rule in conjunction with a new distance metric. The standard Tukey rule is modified to remove the outliers adaptively by considering whether the data is distributed to the left, right or even to the input data's mean value. The elimination of outliers is applied in the proposed modification of the k-means before calculating the centroids to minimize the outliers' influences. Meanwhile, a new distance metric is proposed to assign each data point to the nearest cluster. In this research, the modified k-means significantly improves the clustering accuracy and centroids convergence. Moreover, the proposed distance metric's overall performance outperforms most of the literature distance metrics. This manuscript's presented work demonstrates the significance of the proposed technique to improve the overall clustering accuracy up to 80.57% on nine standard multivariate datasets.

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1. Introduction

Data clustering is a widespread technique for data analysis to categorize underlying patterns into a set of groups or clusters. Each cluster consists of similar data that make it widely different from the other ones. The clustering process simplifies the modelling of data discovery and widely used in various applications such as image segmentation (Siddiqui et al., 2013), bioinformatics (Uma Maheswari, 2020) and nondestructive testing (Shrifan et al., 2021). Recently, data clustering is used to analyze and understand the distribution of coronavirus (COVID-19) in Italy (Doroshenko, 2020).

Various approaches are used for data clustering including partitioning-based methods, hierarchical-based methods and density-based methods. The partitional clustering algorithms such as fuzzy c-means (FCM) (Bezdek et al., 1981) and farthest first algorithm (FFA) (Sharmila, 2013) define a known number of clusters and assign each data object to the nearest centroid based on a distance metric. In the hierarchical-based methods including hierarchical clustering algorithm (HCA) (Hochbaum and Shmoys, 1985) and clustering using representatives (CURE) (Guha et al., 1998), a tree structure of clusters is established without any prior knowledge about the number of clusters. Thereafter, a set of data points are merged into more general clusters or to be divided into smaller clusters sequentially. In the density-based clustering, data clusters are constructed based on the density of data points in a given radius that includes a minimum number of instances such as density-based spatial clustering of applications with noise (DBSCAN) (Ester et al., 1996) and distribution-based clustering of large spatial databases (DBCLASD) (Xiaowei et al., 1998). Considering the wide range of clustering algorithms, it is difficult to assign a particular algorithm as significantly better than another one (Aggarwal and Reddy, 2013). The choice among them is depending on the dataset's properties and the input parameters (Govindaraju et al., 2018).

On other hand, data clustering based on optimization algorithms is one of the most robust techniques due to its capability of discovering the optimal form of data clusterings such as genetic algorithm (GA) (Maulik and Bandyopadhyay, 2000), combinatorial particle swarm optimization (CPSO) (Jarboui et al., 2007) and multi-verse optimizer (MVO) (Shukri et al., 2018). The superiority of the

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Peer review under responsibility of King Saud University.

Production and hosting by Elsevier

https://doi.org/10.1016/j.jksuci.2021.07.003
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optimization techniques in term of clustering accuracy positively overcomes numerous conventional clustering techniques. However, these algorithms still suffer from being trapped into local extrema (Zhang et al., 2011) and generating redundant solutions (Benmessahel and Touahria, 2010). Moreover, many unsupervised clustering techniques are proposed based on blind source separation techniques which a set of observations are modelled as linear combinations to be statistically separated into independent sources or components such as independent component analysis (ICA) (Safont et al., 2017), principal component analysis (PCA) (Kaya et al., 2017)and non-negative matrix factorization (NNFM) (Laxmi Lydia et al., 2020). Although the encouraging performance of data clustering-based source separation techniques on linear data, these algorithms perform poorly due to the outlier presence degrades the data linearity and clustering accuracy (Lever et al., 2017).

Among the clustering algorithms, k-means (MacQueen, 1967) is one of the most popular partitioning algorithms and it used intensively due to its efficiency and simplicity of implementation (Xie et al., 2019). However, it performs poorly due to the presence of outliers which influence the clustering quality (Mousavi et al., 2020). According to Hawkins (Hawkins, 1980); " *An outlier is an observation which deviates so much from other observations as to arouse suspicions that a different mechanism generated it*". Naturally, the real-world data is not always ideal and may include noisy data due to abnormal phenomena or a different measurement mechanism that produces outliers. Recently, outliers processing became a trending topic among the developers of data mining techniques (Gupta and Chandra, 2020). Therefore, it is worthy of considering the impacts of outliers once modifying the k-means algorithm.

Besides the influence of outliers, a different similarity metric leads to different clustering form that may increase or decrease the k-means clustering accuracy (Gupta and Chandra, 2020). Among the distance metrics, Euclidean distance is commonly used with k-means for data clustering. Also, cosine and correlation are the most well-known metrics for clusters differentiation. Clustering the same dataset based on these metrics may produce various manners of clustering which highly depends on the distance model that fits the data domain (Bekhet and Ahmed, 2020). Evaluation of the impact of various similarity metrics can be found in Gupta and Chandra (2020), Gupta and Chandra (2020) and Singh et al. (2013) while selecting or develop suitable distance measure is still an active field among the researchers in the data clustering community.

In this paper, a modified k-means algorithm is proposed for improving data clustering while both online clustering and big data clustering are out of this work's scope. The modified algorithm aims to mitigate the outliers' influences during the centroid measurement for better clustering accuracy. In the proposed algorithm, a well known Tukey's rule is adapted and applied instead of distance metric to identify the boundaries of the outliers-free data. After that, each cluster's centroid is measured based on the average of upper and lower boundaries of the outliers-free data. The average measurement is applied to each attribute of the data points independently. This eliminates the influences of the outliers without removing the entire data point. On the other hand, a new distance metric is proposed in this research. Initially, the ratio is computed between every two corresponding values in the compared attribute. Then, the angle of each ratio is calculated using arctangent. The difference between the produced angles is measured for each attribute in the compared vectors. Finally, root mean square error (RMSE) of the angles' differences is computed as a similarity metric. The proposed method has improved the overall clustering accuracy up to 80.57% on nine standard multivariate datasets compared to the literature algorithms. In this paper, the related work for improving the clustering accuracy of k-means from the perspective of outliers removal and distance metrics is presented in Section 2. The discussion of the conventional k-means, various distance metrics and Tukey's rule for outliers removal is presented in Section 3. The proposed method including the new distance metric, the adapted Tukey's rule and the modified k-means are introduced in Section 4. Sections 5 explains the results and discussion while the challenges and limitations are addressed in Section 6. Finally, the conclusion and future work is summarized in Section 7.

2. Related work

Many studies dealt with improving the clustering accuracy of the k-means algorithm based on various techniques of outliers removal. In term of outliers detection-based distance metric, several studies have recognized the outliers based on the distance between the data point and its closest centroid (Sarvani et al., 2019; Barai (Deb) and Dey, 2017). In these techniques, the data point with a larger distance to the nearest centroid is recognized as an outlier. Additionally, the data points with both low density and large distance to their centroids are considered as outliers as presented in He et al. (2020). In a different approach, local search techniques (Gupta et al., 2017; Friggstad et al., 2019) are used to assist the k-means for outliers detection. The local search aims to remove a few data points from the data within the cluster for minimizing the objective function. If the removed data points have minimized the objective function then those data points are considered as outliers and grouped in a separate cluster. In term of preprocessing techniques, k-means++ is utilized as an additional filtering step in Im et al. (2020) to remove out z of data points as outliers before applying the conventional k-means. Although, the encouraging clustering results of these techniques, the clustering process was only performed on the remaining data which is outlier-free. The outliers data are completely removed and not classified to any known cluster as collected initially.

In other studies, outlier detection is used as an advantage to separate an object from its background such as in image processing (Tu et al., 2020, 2019). However, few studies dealt with mitigating outliers' effects from the mean measurement and classifying all data points into known clusters as collected initially. In Olukanmi et al. (2017), a k-means# is proposed to eliminate the outliers' influences from the clusters' centroid. The detected outliers are completely excluded from the mean measurement only but they are involved later in the clustering process. Thus, the effect of the outliers is mitigated from the centroid measurement and enhanced the clustering accuracy. Although the proposed technique outperformed the conventional k-means, the data point with N attributes was eliminated completely from centroid measurement. In this case, the algorithm cannot recognize an outlier's presence in every attribute independently. This is because the single value of the distance metric represents the entire vector instead the single attribute be removed. Therefore, an empty cluster may occur in case of the presence of at least one outlier in each data point.

Improvement of the clustering accuracy from the perspective of distance metric is demonstrated in various studies. Probabilistic distance for ICA mixture models (PDI) is proposed in Safont et al. (2018). The distance measures the discordance between the probability density of the data, especially to the parameters of each ICAMM model. The source separation of ICAMM is improved based on PDI distance especially after adjusting a threshold value. Although, the good performance for detecting the flaws and the variation in electroencephalographic (EEG). The authors have suggested parallel computing for reducing the processing time.

In Meng et al. (2018), several orders of derivative information is measured between the compared vectors and added to the distance metric. The added information of the derivatives is useful

for capturing the differences between the compared functional data. However, this technique is computational complex due to the calculation of several derivative orders of the functional data. In term of hybrid distance metrics which is commonly used for improving clustering accuracy, a new distance metric named "direction-aware" is developed in Gu et al. (2017) to improve the clustering accuracy of k-means. The proposed distance combines the conventional Euclidean distance to handle the spatial similarity while the cosine metric calculates the shape similarity. Compared to the original metrics, the hybridization of both metrics in a single one has improved the clustering purity. Moreover, a weighted sum of the Euclidean and Pearson distance is introduced in Immink and Weber (2015) using a weight for summation both of the Euclidean and Pearson coefficients. The hybrid distance has improved the similarity between the compared signals once the noise is added significantly.

The clustering accuracy can be improved as long as the outlier is removed before measuring the cluster's centroid as discussed in this section. In this study, a well-known approach such as Tukey's rule for outlier removal is adapted to be used instead of the distance metric. Moreover, the hybridization approach of several distance metrics is followed due to its efficiency for improving the clustering accuracy as discussed earlier. Taking the advantages of commonly used distance metrics such as Euclidean, cosine and correlation into a new single similarity metric can process the data from different perspectives. The capability of hybrid distance approaches for improving the clustering accuracy is high especially after eliminating the influence of the outliers from the centroid of data clusters. The theoretical background of the used techniques such as k-means, Tukey's rule and similarity metrics are discussed in the next section.

3. Theoretical background

3.1. K-means clustering ALGORITHM

The approach of k-means is based on spherical clusters in which the data points converge surrounding the cluster's centroid. The k-means splits a set of data points $X = \{x_1, x_2, x_3, \cdots, x_N\}$ into k known number of clusters. Randomly, the k-means selects k set of centroids $C = \{c_1, c_2, c_3, \cdots, c_k\}$ where$k \leq N$. Thereafter, each data point $x_i$is assigned to the nearest cluster C_j based on the smallest Euclidean distance. The mean of data points μ_j within each cluster is computed to update the centroid in each iteration. The procedure is repeated until there are no changes in the centroid values or the number of maximum iterations is reached. The similarity of the data points within each cluster is high through minimizing the distance among them as given in (1).

$$\underset{C}{arg\,min} \left\{ \sum_{j=1}^{k} \sum_{X \in C_j} \left| X - \mu_j \right|^2 \right\} \tag{1}$$

$$\mu_j = \frac{1}{N} \sum_{X \in C_j} X \tag{2}$$

where *argmin* is the objective function of the k-means. The symbol $|\cdot|$ denotes to the Euclidean distance while N denotes the number of the data points within j^{th} cluster.

3.2. Tukey's rule for outliers removal

Tukey's rule is one of the most robust used techniques for outliers' detection in multivariate data (Huyghues-Beaufond et al., 2020). Tukey defines outliers as the data located outside the boundaries of the inner fence. To detect the outliers, let a set of data points $X = \{x_1, x_2, x_3, \cdots, x_N\}$ includes some outliers where $X \in \mathbb{R}$ and $x_1 \leq x_2 \leq x_3 \leq \cdots \leq x_n$. In the beginning, the first quartile $Q1$ and third quartile $Q3$are computed as stated in (3) and (4) respectively. Thereafter, the inter-quartile range IQR is measured using (5). Finally, the mutation of lower bound lb and upper bound ub are given in (6) and (7) respectively. Therefore, any data point x_i is lower than lb or greater than ub is considered as an outlier.

$$Q1 = x_i | i = round((N+1) \times 0.25) \tag{3}$$

$$Q3 = x_i | i = round((N+1) \times 0.75) \tag{4}$$

$$IQR = Q3 - Q1 \tag{5}$$

$$lb = Q1 - r \times IQR \tag{6}$$

$$ub = Q1 + r \times IQR \tag{7}$$

where i is the data point index while Ndenotes the total number of data points. r is a predefined constant to adjust the inliers' boundaries. Tukey proposed a predefined constant $r = 1.5$. However, there is no statistical basis for the reasoning of the proposed value (Seo et al., 2006). Therefore, this value can be adapted based on its application.

3.3. Similarity measurements

Euclidean distance is the most common similarity metric among various machine learning algorithms (Mesquita et al., 2017). For measuring the distance between two vectors, let $A = \{a_1, a_2, a_3, \cdots, a_N\}$ and $B = \{b_1, b_2, b_3, \cdots, b_N\}$ are two data points with N numeric attributes. The Euclidean distance d_E between A and B can be measured as given in (8).

$$d_E(A, B) = \sum_{i=1}^{N} |a_i - b_i| \tag{8}$$

Several properties are satisfied by Euclidean distance such as non-negative where $d_E \geq 0$, once both A and B are completely similars the distance $d_E = 0$ and the distance is a symmetric function where $d_E(A, B) = d_E(B, A)$ (Han et al., 2012). Despite the encouraging performance in the categorical and numerical datasets, the performance of the Euclidean distance is relatively poor for mixed datasets (Hu et al., 2016).

On the other hand, cosine metric computes the similarity between two vectors based on the inner product as defined in (9) (Lo et al., 2018). Cosine distance $d_C(A, B) = 1$ indicates that both vectors have no similarity. The higher similarity between the input vectors is once the cosine distance tends to 0.

$$d_C(A, B) = 1 - \frac{\sum_{i=1}^{N} a_i b_i}{\sum_{i=1}^{N} a_i^2 \sum_{i=1}^{N} b_i^2} \tag{9}$$

Typically, the cosine distance is utilized to measure the text-similarity in the document classification (Manning et al., 2008). Cosine distance performs well on the categorical and numerical datasets independently and worst in the mixed datasets (Hu et al., 2016).

In a different approach, Pearson coefficient is utilized to compute the correlation degree between A and B data points on the straight regression line (Yin and Wang, 2020). The similarity metric is defined by subtracting the Pearson correlation coefficient from 1 as stated in (10).

$$d_R(A, B) = 1 - \frac{\sum_{i=1}^{N} (a_i - \overline{A})(b_i - \overline{B})}{\sqrt{\sum_{i=1}^{N} (a_i - \overline{A})^2} \sqrt{\sum_{i=1}^{N} ((b_i - \overline{B})^2}} \tag{10}$$

where $\overline{A}$ and $\overline{B}$ are the means of A and B vectors, respectively. Both data scaling and localization are invariant to the Pearson correlation coefficient. Thus it is not a distance metric since the approach aims to define the variations in the data's shape instead of calculating the magnitude differences.

4. Proposed method

4.1. Distance measurement

In this study, a new similarity metric is proposed to replace the Euclidean distance in the conventional k-means. The proposed similarity metric aims to take the advantages of various measurements such as Euclidean distance, cosine metric and correlation coefficient due to a different distance metric leading to a different manner of the clustering accuracy (Aggarwal et al., 2019).

Let $A = \{a_1, a_2, a_3, \cdots, a_N\}$ and $B = \{b_1, b_2, b_3, \cdots, b_N\}$ are two vectors with N numeric attributes. To measure the similarity between A and$B$, a normalization process is firstly applied to convert them into positive values between 1 and 90 as follow:

$$Y = \max(A, B) \tag{11}$$

$$Z = \min(A, B) \tag{12}$$

$$A = \frac{A - Z}{Y - Z} \times (r_{max} - r_{min}) + r_{min} \tag{13}$$

$$B = \frac{B - Z}{Y - Z} \times (r_{max} - r_{min}) + r_{min} \tag{14}$$

where $Y$and$Z$denote the maximum and minimum values among A and B respectively.$r_{max}$and r_{min}denote the maximum and minimum of the target scaling which are 90 and 1 respectively.

The normalization process aims to guarantee both values of a_i and b_i is not equal to zero for avoiding the division by zero in (15) and (16). After the normalization, the proportionality between the normalized vectors A and$B$ is computed firstly to indicate the number of times for one value is contained in the other one. The proportionality can be computed by measuring the ratios between every two corresponding values such as a_i/b_i andb_i/a_iin the compared vectors A and$B$. In the case of fully similar vectors, both ratio coefficients equal 1 while a different value is produced in dissimilarity.

After that, the angles α_i and $\beta_i$of both ratios are computed as given in (15) and (16). For similar vectors, both angles tend to 45° while the angles of the dissimilar values tend between 0° and 90° as shown in Fig. 1. Therefore, the distance between α_i and β_i is decreased while the difference between them is small and vice versa. As a result, N of angles' differences are generated for the compared vectors A and Busing$|\alpha_i - \beta_i|$. Finally, root mean square error (RMSE) of the angles' differences is computed as stated in (17). RMSE tends to 0 in the case of similar vectors and a large value for the dissimilar vectors. In this technique, RMSE is the last outcome of the distance measurement. RMSE is not measured from the raw data directly, but it is computed from the angles' differences of the ratio coefficients.

$$\alpha_i = \tan^{-1}\frac{a_i}{b_i}, \text{ where } b_i \neq 0 \tag{15}$$

$$\beta_i = \tan^{-1}\frac{b_i}{a_i}, \text{ where } a_i \neq 0 \tag{16}$$

$$d_P(A, B) = \sqrt{\frac{1}{N}\sum_{i=1}^{N}|\alpha_i - \beta_i|^2} \tag{17}$$

The proposed distance d_P satisfies several mathematical properties such as non-negative where $d_P \geq 0$. In the case of both A and B are completely similar, the distance $d_P = 0$. Moreover, the proposed distance is a symmetric function if $d_P(A, B) = d_P(B, A)$. Besides, several advantages are considered while computing the proposed distance. The advantages of the correlation distance are used at the first stage by computing the proportionality between the compared vectors. Secondly, the compared vectors' angle metric is utilized using the arctangent function, which is similar to the cosine metric. Thirdly, the angles difference $|\alpha_i - \beta_i|$ is achieved similarly to the Euclidean distance. Finally, RMSE is used to produce the last outcome of the proposed distance d_P The value of d_P tends to 0 if there is a strong correlation between the compared vectors due to the entire points lie on the regression line. Thus, the proposed distance metric is computed to be used with the modified k-means as discussed in Section 4.3.

4.2. Adapted Tukey's rule

The standard Tukey's rule as discussed in Section 3.2 aims to remove the outliers using Tukey's constant $r = 1.5$in both lower and upper boundaries equally. Meanwhile, the distribution of the remaining data concerning their mean is not considered, including skewness. The skewed data means that the data is not equally distributed along the left and right sides of the mean value. Therefore, computing the mean from the remaining data after applying Tukey's rule does not make sense while the hard skewness has existed. The skewed data influences the centroid measurement of the k-means. Therefore, an adaption to the standard Tukey's rule is proposed in this paper to provide a robust measurement of the

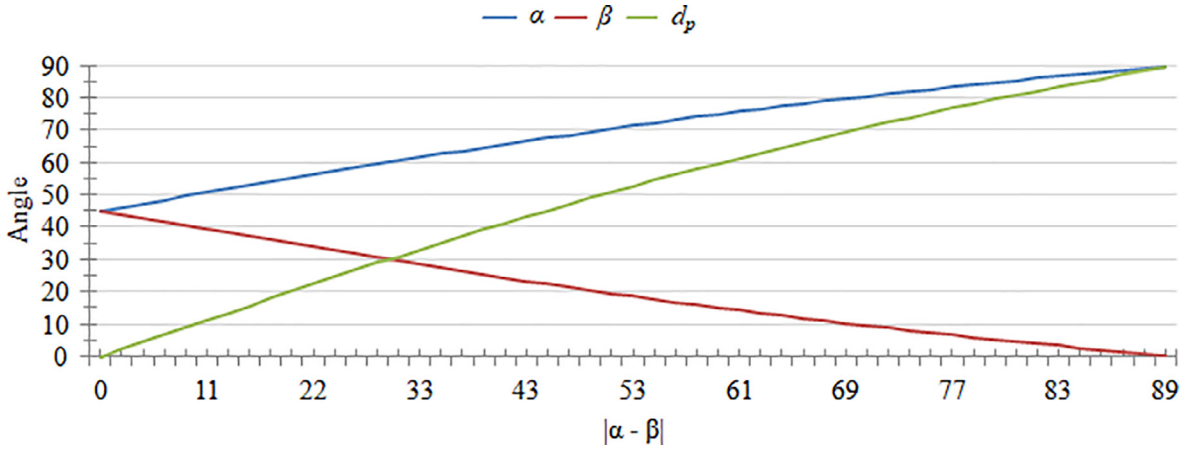


Fig. 1. The relationship between the angles differences $|\alpha - \beta|$ and the proposed distance metricd_P.

k-means' centroids by mitigating the data skewness through outliers removal. This provides outliers-free data which does not influence the centroid measurement.

Typically, the skewness of the distributed data can be detected by counting the number of data points along the mean value's right and left sides. Therefore, skewness exists if the number of the data points is not balanced equally along the left and the right sides of the mean value. Otherwise, the data points are normally distributed. The Tukey's constant r is divided into r_1 and r_2 to manage the amount of the removed outliers at each side of the distributed data where $r_1 > r_2$ (e.g., $r_1 = 0.85$ and $r_2 = 0.65$). In the case of skewed data, r_1 allows removing fewer data points due to its large value which increases the boundary of the data to be involved at the skewness position. Conversely, r_2 allows removing a large number of the data points at the boundary away from the skewness position. The small value of r_2 decreases the data boundary to be involved far from the skewness position. On the other hand, the amount of the removed data is set into the average of r_1 and r_2 in case of the data is equally distributed along both sides of the mean value. The formulation of the adaptive process is given in (18) and (19).

$$lb = \begin{cases} Q1 - r_1 \times IQR & | \quad GM < LM \\ Q1 - r_2 \times IQR & | \quad GM > LM \\ Q1 - \frac{r_1 + r_2}{2} \times IQR & | \quad GM = LM \end{cases} \tag{18}$$

$$ub = \begin{cases} Q3 + r_2 \times IQR & | \quad GM < LM \\ Q3 + r_1 \times IQR & | \quad GM > LM \\ Q3 + \frac{r_1 + r_2}{2} \times IQR & | \quad GM = LM \end{cases} \tag{19}$$

where GM and LM are the numbers of the data points that greater and less than the mean value respectively.

The modified Tukey's rule aims to adaptively eliminate the outliers concerning three data distribution cases, whether the data is distributed at the left, right or equally to the mean value. This is useful to decide where the hard and less removal should be applied instead of eliminating the outliers from both sides equally as proposed in the standard Tukey's rule. The purpose of adapting Tukey's rule is to provide robust outliers-free boundaries. These boundaries such as lb and ub can be used to compute the clusters' centroids of the modified k-means only. Outliers removal is only applied during the centroids measurement. Therefore, none of the input data will be removed and all of them will be clustered to the known group as initially collected. A detailed discussion about the hybridization between the adapted Tukey's rule and k-mean is introduced in the next section.

4.3. Modified k-means

In this paper, the modification of the k-means aims to mitigate influences of the outliers while computing the clusters' centroids. Let X be a numerical $N \times M$ dataset where N denotes the number of data points and M is the number of attributes. Algorithm 1 shows the pseudocode of the modified k-means and the entire steps are discussed as the following:

1. *Initialization:* The modified k-means selects k set of the data points randomly as data centroids $C = \{c_1, c_2, c_3, \cdots, c_k\}$ where $k \leq N$ and $C \in X$.
2. *Assignment*: each data point $x_i$is assigned to the nearest cluster c_j based on the proposed distance metric as given in Eq. (17). At this step, k sets of the data clusters are constructed initially.
3. *Outliers' removal*: the adapted Tukey's rule is applied to each cluster for removing the outliers in each m attribute independently. The adapted Tukey's rule is applied recursively to each attribute's remaining data until no more outliers are detected.
4. *Centroid updating:* the new centroid c_j of each attribute m within-cluster j is computed as the following:

$$c_{j,m} = \frac{lb_m + ub_m}{2} \tag{20}$$

where lb and ub are the lower and upper boundaries obtained from (18) and (19).

5. Termination criteria: stop if all data centroids in C are not changed or maximum iteration is reached. Otherwise, repeat the steps from 2 to 5.

Applying the adapted Tukey's rule recursively has two objectives. The first one is to obtain the outliers-free data's boundaries by mining the underlying patterns where the data within each cluster is focused. Therefore, the recursive process helps remove the hidden outliers that other outliers may cover. The second one is to remove the outliers from each attribute independently instead of removing the entire observation as commonly used by the clustering algorithms based on outliers' removal techniques. Processing each attribute independently is useful to avoid the occurrence of the empty cluster once at least one outlier is detected in each observation.

Algorithm 1: The modified k-means clustering algorithm.

Input: X is an input dataset with$N$ data points and M attributes. k is a predefined clusters' number. *MaxIter* is a maximum number of iterations.
Output: $S$is a set of k clusters belong to X dataset.

1. **Begin**
2. *// Initialization*
3. Select random k of initial centroids $C = \{c_1, c_2, c_3, \cdots, c_k\}$ where $C \in X$.
4. **Repeat until** *MaxIter* is met **or** C is not changed *// Termination criteria*
5. $S = Null$
6. *// Assignment*
7. **For** $i = 1 : N$
8. **For** $j = 1 : k$
9. $d(j) = d_p(x_i, c_j)$, compute the distance between $x_i$and $c_j$as given in (17).
10. $S_j = x_i$, assign x_i to its cluster setS_j if $x_i$has a minimum $d(j)$ withc_j.
11. *// Outliers' removal*
12. **For** $j = 1 : k$
13. **For** $m = 1 : M$
14. **Repeat until** no outlier is detected in $S_j(:, m)$ *// ":" denotes to all values*
15. Compute $lb$and ub of $S_j(:, m)$as given in (18) and (19) respectively.
16. **For** $i = 1 : length(S_j(:, m))$
17. **If** $S_j(i, m) < lb$ **or** $S_j(i, m) > ub$ *// Check if an outlier*
18. Remove the outlier data point$S_j(i, m)$.
19. *// Centroid updating*
20. $c_j(m) = (lb + ub)/2$, update the centroid of each attribute as given in (20).
21. **End**

In the modified k-means, the centroid is measured from the average of *lb* and *ub* of the outliers-free data at the last iteration of the adapted Tukey's rule. This step replaces the traditional centroid measurement μ_j in the conventional k-means. This step's replacement allows the modified k-means to quickly calculate the centroid from the outlier-free data's boundaries instead of computing the mean of entire input data. On the other hand, the modified k-means allows the entire data points to be clustered based on the proposed distance metric. None of the data points will be removed from the input data like usually used in the conventional clustering algorithms based on outliers removal methods. The proposed method is validated on various commonly used datasets as discussed in the next section.

5. Results and discussion

5.1. Datasets

The proposed method is validated on nine multivariate datasets obtained from the machine learning repository of the University of California, Irvine (UCI) (Dua et al., 2020). The used datasets have a diverse number of attributes ranged from 4 to 13. Moreover, the number of clusters of the given datasets is varied from 2 to 6. The content of the used datasets is described and summarized in Table 1.

5.2. Evaluation of the proposed method

5.2.1. Evaluation of the clustering accuracy of the modified k-means

The modified k-means is evaluated on the UCI as mentioned earlier datasets. The given datasets are normalized by ranging their values from 1 to 90 to avoid dividing by zero of the given equations in (15) and (16). The obtained results are compared to the common literature algorithms such as k-means (KMN), hierarchal clustering algorithm (HCA) and farthest first algorithm (FFA). Moreover, a comparison has been introduced between the proposed algorithm and well-known evolutionary algorithms such as clustering-based PSO (CPSO), clustering-based GA (CGA) and static clustering-based MVO (SCMVO). Also, the results of modified k-means are compared to k-means# (KMN#) and DBSCAN which are robust algorithms to outliers' influences. In the experimental setup, the constant values of the adapted Tukey's rule r_1 and r_2 are set up into 0.85 and 0.65 respectively as given in Eqs. (18) and (19) where $r_1 > r_2$. These constants are empirically selected based on the best clustering accuracy and globally used for all datasets during this experiment. The utilized distance measurement of the proposed method is given in (17). The maximum number of iterations of the modified k-means is 50 while the accuracy results are obtained from the mean value over 100 runs. The clustering accuracy in each run is measured based on the clustering purity which is the percentage of the number of data points that are correctly clustered as given in (21).

$$\text{Purity} = \frac{1}{N}\sum_{i=1}^{k} |L_i \cap C_i| \tag{21}$$

where *N*denotes the total number of data points. $L_i$denotes the number of true data points within each cluster *i*. $C_i$denotes the number of data points that are correctly clustered using a clustering algorithm.

The proposed method overcomes most of the literature algorithms as shown in Table 2. The modified k-means significantly outperformed the conventional k-means, k-means# and the evolutionary algorithms such as CPSO, CGA and SCMVO in all datasets especially Glass, Balance, Vertebral, Ecoli and Blood. However, HCA overcomes the proposed algorithm in the Balance dataset with 2.55% of accuracy difference. Meanwhile, both HCA and FFA outperform the modified k-means in the Blood dataset with a 0.47% accuracy difference, which is a competitive performance. Moreover, DBSCAN achieves better performance in the Blood dataset compared to the proposed method with 2% of accuracy difference.

The modified k-means achieves a significant performance for the most challenging databases with a large number of classes such as Glass and Ecoli with 6 and 5 classes, respectively. Although the initial centroids randomly selected, the low rate of the standard deviation over 100 runs reflects the stability as well as the robustness of the proposed technique. Generally, the overall performance of the modified k-means performed the best accuracy results in a total of 7 datasets and competitive accuracy results in the other 2 datasets. As a result, the modified k-means completely overcomes 5 well-known algorithms, and with competitive results compared to the other three in the two datasets.

5.2.2. Impact of outliers removal on the clustering accuracy

The impacts of both standard and adapted Tukey's rules on the modified k-means are discussed in Section 5.2.2.1. Moreover, the mechanism of outliers removal through the clustering process of both outlier techniques is addressed in Section 5.2.2.2. In the experimental setup, the adapted Tukey's rule parameters r_1 and r_2 are set up into 0.85 and 0.65 respectively. In the standard Tukey's rule, Tukey's constant $r = 1.5$.

5.2.2.1. Clustering accuracy through the modified k-means' iterations. Fig. 2 shows the accuracy variations during the iteration incremental of the modified k-means based on the adapted Tukey's rule. Each iteration has a valuable contribution in increasing the clustering accuracy on most datasets as shown in Fig. 2. However, the loss in the accuracy during the iteration increment can be seen on the Blood dataset, which inferences the modified algorithm's performance compared to the HCA and FFA. This behaviour of

Table 1
The description of the used datasets obtained from UCI repository.

Dataset	No. of attributes	No. of classes	No. of data points	No. of data points in each class	Data type
Iris	4	3	150	50, 50, 50	Real
Glass	9	6	214	70, 17, 9, 76, 29, 13	Real
Balance	4	3	625	49, 288, 288	Real
Cancer	8	2	699	458, 241	Real
Wine	13	3	178	59, 71, 48	Real
Vertebral	6	2	310	207, 100	Real
Ecoli	7	5	327	143, 77, 35, 20, 52	Real
Blood	4	2	748	570, 178	Real
Seed	7	3	210	70, 70, 70	Real

Table 2
Accuracy results of the modified k-means compared to other literature clustering algorithms. The bold values in all tables refer to the best performance.

Dataset	KMN (MacQueen, 1967) Mean ± Std	HCA (Hochbaum and Shmoys, 1985) Mean ± Std	FFA (Sharmila, 2013) Mean ± Std	CPSO (Jarboui et al., 2007) Mean ± Std	CGA (Maulik and Bandyopadhyay, 2000) Mean ± Std	CSMVO (Shukri et al., 2018) Mean ± Std	DBSACN (Ester et al., 1996) Mean ± Std	KMN# (Olukanmi et al., 2017) Mean ± Std	PROPOSED Mean ± Std
Iris	0.57 ± 0.24	0.89 ± 0.00	0.86 ± 0.00	0.96 ± 0.00	0.96 ± 0.00	0.96 ± 0.00	0.68 ± 0.00	0.85 ± 0.10	**0.96 ± 0.00**
Glass	0.36 ± 0.08	0.46 ± 0.00	0.48 ± 0.00	0.45 ± 0.00	0.36 ± 0.00	0.52 ± 0.08	0.46 ± 0.00	0.49 ± 0.04	**0.56 ± 0.03**
Balance	0.47 ± 0.17	0.63 ± 0.00	**0.65 ± 0.00**	0.37 ± 0.18	0.41 ± 0.18	0.49 ± 0.17	0.47 ± 0.00	0.55 ± 0.07	0.61 ± 0.07
Cancer	0.95 ± 0.00	0.66 ± 0.00	0.84 ± 0.00	0.96 ± 0.00	0.96 ± 0.00	0.96 ± 0.00	0.93 ± 0.00	0.96 ± 0.03	**0.97 ± 0.00**
Wine	0.90 ± 0.14	0.40 ± 0.00	0.70 ± 0.00	0.91 ± 0.12	0.95 ± 0.01	0.96 ± 0.00	0.61 ± 0.00	0.71 ± 0.04	**0.98 ± 0.00**
Vertebral	0.68 ± 0.00	0.67 ± 0.00	0.68 ± 0.00	0.68 ± 0.02	0.70 ± 0.00	0.71 ± 0.00	0.67 ± 0.00	0.73 ± 0.04	**0.78 ± 0.01**
Ecoli	0.66 ± 0.11	0.65 ± 0.00	0.60 ± 0.00	0.60 ± 0.11	0.62 ± 0.13	0.57 ± 0.13	0.43 ± 0.00	0.66 ± 0.09	**0.71 ± 0.07**
Blood	0.53 ± 0.06	0.76 ± 0.00	0.76 ± 0.00	0.48 ± 0.00	0.48 ± 0.00	0.48 ± 0.01	**0.78 ± 0.00**	0.68 ± 0.03	0.76 ± 0.02
Seed	0.79 ± 0.19	0.90 ± 0.00	0.67 ± 0.00	0.89 ± 0.05	0.90 ± 0.00	0.90 ± 0.00	0.68 ± 0.00	0.90 ± 0.04	**0.94 ± 0.00**
Overall accuracy	0.66 ± 0.11	0.67 ± 0.00	0.69 ± 0.00	0.70 ± 0.05	0.70 ± 0.04	0.73 ± 0.04	0.63 ± 0.00	0.73 ± 0.05	**0.81 ± 0.02**

the proposed algorithm due to the nature of the dataset itself while the variety of the included data could be clustered up to 18 different classes (Lord et al., 2017). Therefore, the high variety of the included data points impacts the accuracy of the modified k-means.

The standard Tukey's rule influences the convergence process due to the number of iterations is increased significantly with less contribution to the accuracy compared to the adapted Tukey's rule as shown in Fig. 3. The adapted Tukey's rule achieves the best performance in term of both iterations' number and accuracy as shown in Table 3. The adapted Tukey's rule positively increases the clustering accuracy while reducing the iterations' number needed for centroids convergence. This improvement is acquired due to the adapted Tukey's rule adaptively eliminates the outliers based on the added rules of removing the outliers as given in (18) and (19). This makes the adapted Tukey's rule quickly reach the boundaries of the outlier-free data where the data centroid should be measured. On the other hand, the modified k-means based on the standard Tukey's rule achieves the best accuracy performance compared to the literature clustering algorithm in four datasets which are Cancer, Wine, Vertebral and Seed as underlined in Table 3.

5.2.2.2. Mechanism of outliers removal. In terms of the outliers removal mechanism, Fig. 4 shows an example of the centroid measurement at each iteration during the outliers removal process. Fig. 4(a) illustrates a description of the boxplot parameters of the adapted Tukey's rule. In this example, the data of the Sepal length attribute of the Iris dataset is monitored in every single cluster during the outliers removal process. In all three clusters, the remaining data at the final iteration is completely outliers-free as shown in Fig. 4(b), (c) and (d). It can be noted that at the first iteration there are some of the outliers were detected and removed from the Sepal length attribute in every cluster. Moreover, there are some of the covered outliers were detected in both Iris Setosa and Iris Virginica clusters after applying the adapted Tukey's rule recursively to the remaining data. Therefore the proposed recursive process of the removal of the outliers is useful to eliminate this kind of outliers. Furthermore, the proposed technique for centroid measurement is different from the centroid of the conventional k-means at the final iteration of removing the outliers. The proposed centroid tends to be close to the mean of the outliers-free data. The proposed centroid is aligned at the middle of the blue box where the data is focused. In term of skewed data, it can be noted that a hard skewness is shown in the Iris Virginica cluster. In this cluster, the data median is shifted at the lower boundary as shown in Fig. 4(d). Therefore, the adapted Tukey's rule mitigates the skewness by eliminated the covered outliers at each iteration. As a result, the centroid is measured after the hard skewness is mitigated at the final iteration. Thus, the adapted Tukey's rule works to provide a robust measurement of the k-means' centroids by mitigating the data skewness through outliers removal. This provides

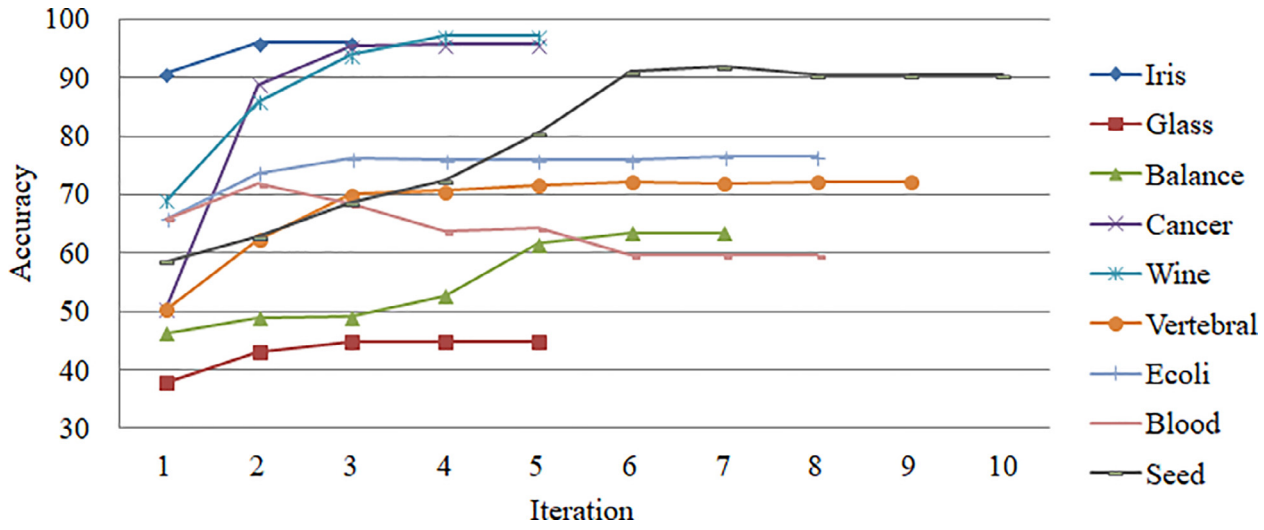


Fig. 2. Evolution of the clustering accuracy through iterations of the modified k-means.

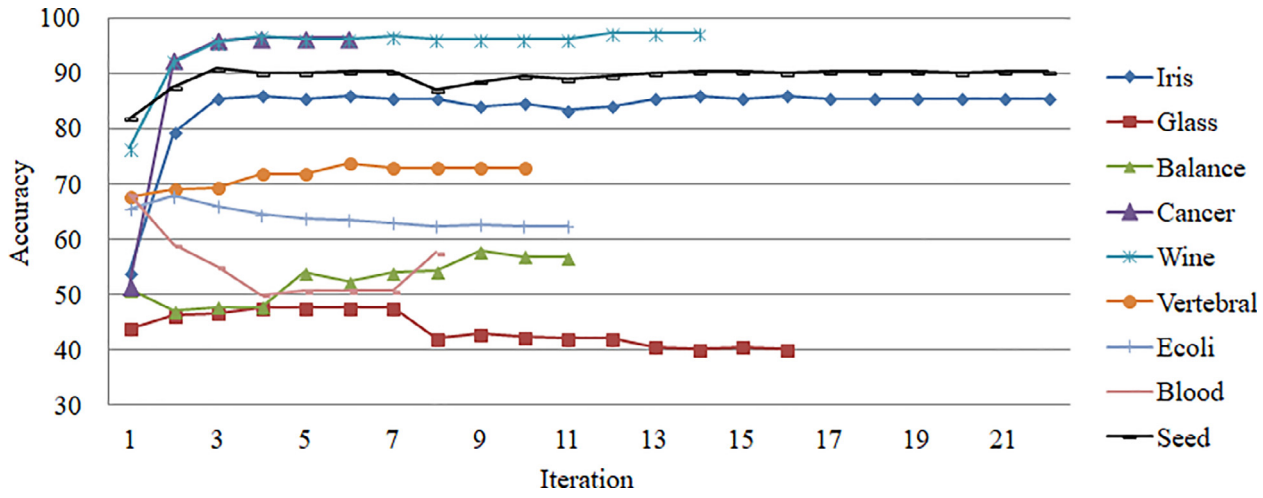


Fig. 3. Evolution of the clustering accuracy through iterations of the modified k-means-based standard Tukey's rule.

Table 3
A comparison between the performance of the standard Tukey's rule and the adapted one in term of iterations' number and clustering accuracy. The bold values in all tables refer to the best performance.

Dataset	Iteration number			Clustering accuracy		
	Standard	Adapted	Difference	Standard	Adapted	Difference
Iris	22	**3**	19	0.8581	**0.9605**	0.1024
Glass	16	**5**	11	0.4558	**0.5593**	0.1035
Balance	11	**7**	4	0.5750	**0.6045**	0.0295
Cancer	6	**5**	1	0.9646	**0.9649**	0.0003
Wine	14	**5**	9	0.9715	**0.9748**	0.0033
Vertebral	10	**9**	1	0.7376	**0.7813**	0.0437
Ecoli	11	**8**	3	0.6518	**0.7113**	0.0595
Blood	**8**	**8**	0	0.6142	**0.7553**	0.1411
Seed	22	**10**	12	0.9075	**0.9392**	0.0317
Average	13.3333	**6.6667**	6.6667	0.7485	**0.8057**	0.0572

outliers-free data which does not influence the centroid measurement.

Compared to the adapted Tukey's rule, the standard Tukey's rule is not sensitive enough to detect the outliers within each cluster of the Iris dataset as shown in Fig. 5. The large value of Tukey's constant $r = 1.5$ maximizes both of lower and upper boundaries. As a result, the covered outliers are not detected because they are quietly placed lower than *ub*and greater than*lb*. Therefore, the computed centroid in this case is similar to that one calculated by the conventional k-means, as illustrated in Fig. 5.

5.2.3. Evaluation of scarce data clustering

The Glass dataset provides a good example of scarce data clustering. In this dataset, six classes are given which the number of data points within each class is 70, 17, 9, 76, 29 and 13 respectively. It is clear that the second, third and fourth classes fall under scarce data clustering due to the small number of data points within these classes compared to the other ones. The proposed method correctly assigns 35.3%, 88.9% and 61.5% of data points in the second, third and fourth clusters respectively. Focusing on the third cluster which only includes nine data points, eight of them are correctly assigned to this cluster. In this cluster, only two data points are wrongly included from other clusters. Considering the clustering complexity of the Glass dataset as shown in Table 2, the proposed method successfully constructs an independent centroid that attracts most of the data points in the second cluster with few data points are brought from other clusters. The performance of the modified k-means is encouraged regarding the scarce data clustering.

5.2.4. Evaluation of clustering run-time

In Table 4, the run-time comparison is addressed among conventional k-means (KMN), k-means-based standard Tukey's rule (KMN-TTR) and k-means-based adapted Tukey's rule (KMN-ATR). The conventional k-means has significantly outperformed both k-means based on standard and adapted Tukey's rules in term of processing time. The additional steps of outlier removal in both algorithms have added extra processing time to the procedure of the conventional algorithm. Compared to the standard Tukey's rule, k-means based on the adapted Tukey's rule is the worst in processing time. In the adapted Tukey's rule, the small values of $r_1 = 0.85$ and $r_2 = 0.65$ have risen the iterations' number of outliers removal. Meanwhile, the large value of Tukey constant$r = 1.5$ has minimized the processing time due to no more outliers are detected and quickly met stop criteria. The given examples in Figs. 4 and 5 clearly illustrate the difference in iterations' number in both rules of outliers removal. Although the added steps by the adapted Tukey's rule have risen the processing time but they significantly improve the clustering accuracy. In some critical applications such as medical requirements, clustering accuracy is essential regardless of the computational time (Walker et al., 2020). Giving priority

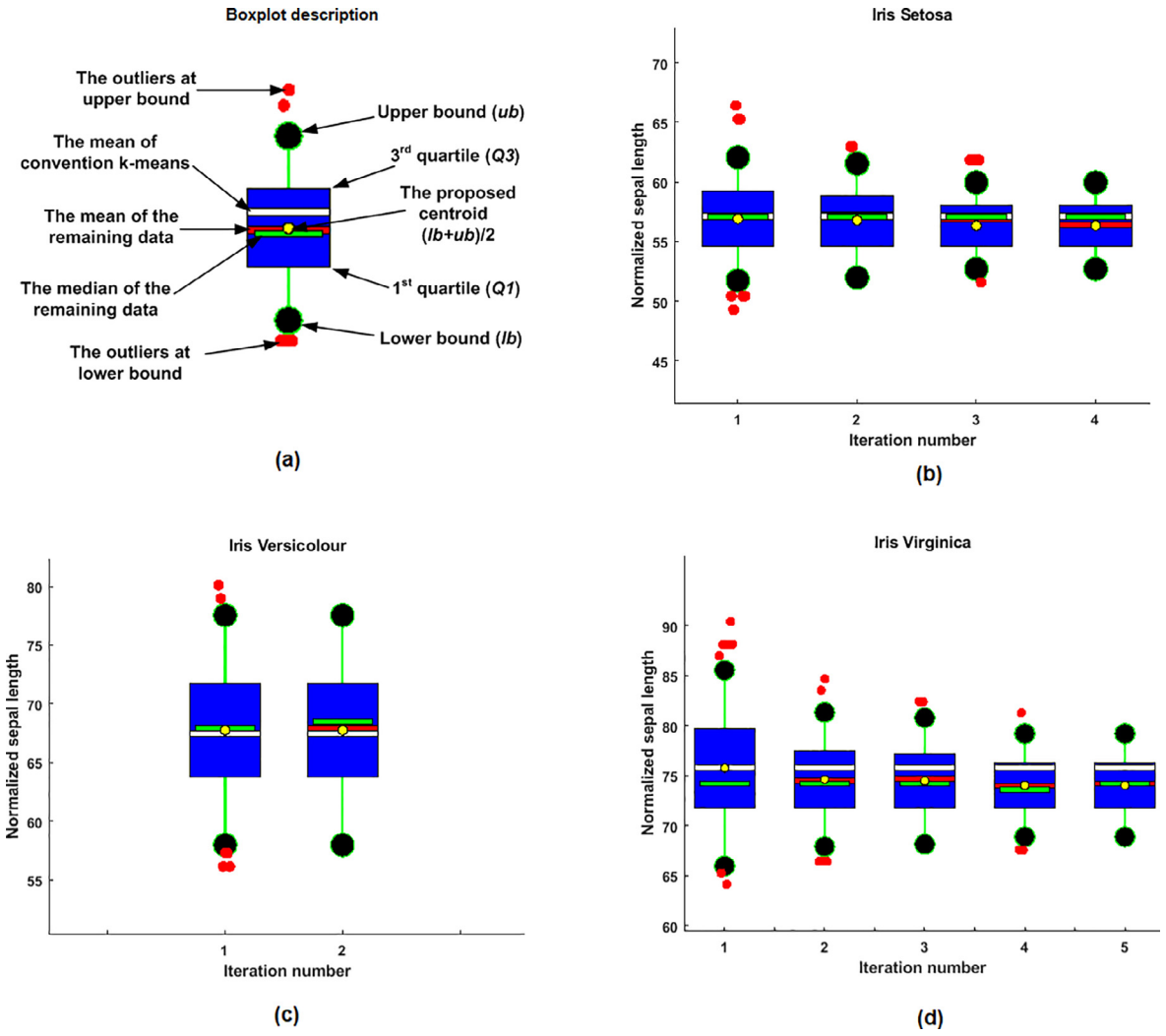


Fig. 4. The centroid measurement of the sepal length attribute within each cluster of the Iris dataset during the outliers removal process using the adapted Tukey's rule.

to the accuracy or time is highly dependent on the used algorithm for such application.

5.2.5. Evaluation of the proposed distance

The clustering accuracy of the proposed distance has outperformed most of the literature distance metrics in various datasets as shown in Table 5. The significant performance of the proposed distance can be observed clearly on Iris, Glass, Wine, Vertebral, Blood and Seed. However, the cosine and correlation metrics only overcome the proposed distance in Balance dataset with 2.95% and 0.85% of accuracy difference, respectively. Moreover, the Euclidean distance overcomes the proposed distance in Ecoli dataset with 3.52% of accuracy difference. The proposed distance achieves very competitive performance with Euclidean distance in the Cancer dataset by 0.09% of the clustering accuracy difference.

On the other hand, the underlined accuracy results mean that the used distance metrics outperformed the entire clustering algorithms stated in Table 2. Therefore, the modified k-means can work adequately to achieve an acceptable accuracy rate once other distance metrics are used such as Euclidean, cosine and correlation. The low STD coefficient illustrates the stability of the modified k-means once working with the utilized distance metrics. Compared to the overall accuracy, the modified k-means achieves the best accuracy performance once various distance metrics are used.

6. Challenges and limitations

In this manuscript, several modifications are proposed for the k-means algorithm based on the adapted Tukey's rule for outliers removal and the new distance metric. Although the significant improvement in clustering accuracy, several challenges face the modified k-means such as real-time clustering due to the outliers removal step that added extra processing time to the procedure of the conventional k-means. Also, the proposed distance metric is limited to real values greater than zero. This needs additional processing time for data normalization. However, the biding between the clustering accuracy and run-time highly depends on the application filed. On the other hand, data clustering with high variety number of possible clusters such as in Blood dataset is another challenge to the proposed algorithm. This variety in number of clusters degrades the clustering accuracy using the modified k-means.

7. Conclusion and future work

This work has demonstrated the significance of the proposed technique to improve the overall clustering accuracy up to 80.57% on nine UCI multivariate datasets. The proposed technique for centroid measurement based on the adapted Tukey's rule sig-

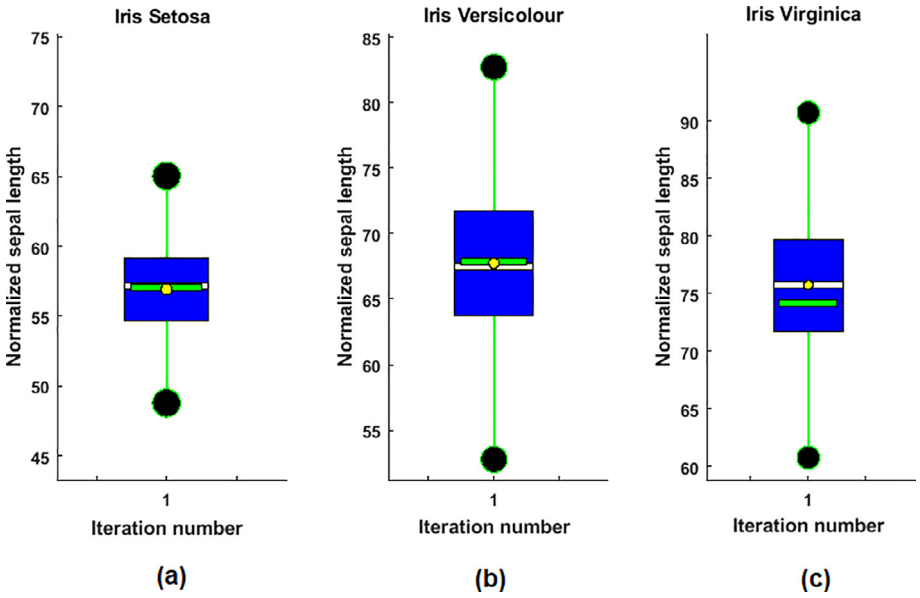


Fig. 5. The centroid measurement of the Sepal length within each cluster of the Iris dataset during the outliers removal process using the standard Tukey's rule.

Table 4
Run-time comparison of KMN, KMN-TTR and KMN-ATR. The bold values in all tables refer to the best performance.

Dataset	Iris	Glass	Balance	Cancer	Wine	Vertebral	Ecoli	Blood	Seed
KMN	**0.0779**	**0.1707**	0.2998	**0.2617**	**0.0901**	**0.0668**	**0.2240**	**0.2775**	**0.1165**
KMN-TTR	0.1027	0.3014	**0.2714**	0.3036	0.1718	0.1618	0.3143	0.3228	0.1606
KMN-ATR	0.1250	0.3448	0.3383	0.2977	0.2321	0.2043	0.3578	0.3342	0.1899

Table 5
A comparison between the accuracy of the proposed distance metric and various literature distance metrics using the modified k-means clustering algorithm. The bold values in all tables refer to the best performance.

Dataset	Euclidean Mean ± STD	Cosine Mean ± STD	Correlation Mean ± STD	Proposed Mean ± STD
Iris	0.9247 ± 0.04	0.8693 ± 0.17	0.8691 ± 0.18	**0.9605 ± 0.00**
Glass	0.5243 ± 0.03	0.5527 ± 0.02	0.5508 ± 0.02	**0.5593 ± 0.03**
Balance	0.5868 ± 0.06	**0.6340 ± 0.05**	**0.6130 ± 0.05**	0.6045 ± 0.07
Cancer	**0.9658 ± 0.00**	0.8531 ± 0.01	0.8430 ± 0.03	0.9649 ± 0.00
Wine	0.9555 ± 0.00	0.8993 ± 0.00	0.9041 ± 0.00	**0.9748 ± 0.00**
Vertebral	0.7251 ± 0.01	0.7615 ± 0.01	0.7394 ± 0.01	**0.7813 ± 0.01**
Ecoli	**0.7465 ± 0.08**	0.6772 ± 0.06	0.6606 ± 0.06	0.7113 ± 0.07
Blood	0.6939 ± 0.03	0.6177 ± 0.07	0.6432 ± 0.05	**0.7553 ± 0.02**
Seed	0.9065 ± 0.13	0.8589 ± 0.02	0.8680 ± 0.00	**0.9392 ± 0.00**
Overall	0.7810 ± 0.04	0.7471 ± 0.05	0.7435 ± 0.04	**0.8057 ± 0.02**

nificantly improved the overall clustering accuracy by 5.72% compared to the standard rule. Moreover, the overall iteration number for centroid convergence is highly reduced by 50% once the adapted Tukey's rule is used. Although the reduction in iteration number, the adapted Tukey's rule has added extra processing time to the procedure of conventional k-means algorithm compared to the standard Tukey's rule.

On the other hand, the developed distance metric properly aided the modified k-means in term of the overall accuracy performance compared to the other distance metrics. Generally, the proposed distance metric outperformed the Euclidean, cosine and correlation metrics by 2.40%, 5.79% and 6.15% of the clustering accuracy, respectively. In future work, the adapted Tukey's rule and the new distance metric will be applied to various clustering algorithms such as fuzzy c-means and k-medoids. Moreover, improving the run-time of modified k-means for big data clustering will be investigated further in future work by reducing the time complexity of Tukey's rule for outlier removal.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This research is partially supported by Fundamental Research Grant Scheme (FRGS), Ministry of Higher Education, Malaysia entitled 'Enhanced Differential Evolution Algorithm by Balancing the Exploitation and Exploration Search Behavior for Data Clustering' account number 203/PELECT/6071398.

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